

RATIONAL SPECIAL VALUES OF WEIGHT-5 CM NEWFORMS: A HIERARCHY AND CLASS-FIELD-THEORETIC STRUCTURE

KÉVIN RÉMONDIÈRE

ABSTRACT. For each imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ of class number 1 (the nine Heegner discriminants $D \in \{-3, -4, -7, -11, -19, -43, -67, -163\}$ together with the twin-pair partners for $D \in \{-11, -19\}$), there is a unique rational CM-by- χ_D weight-5 newform f_D . We define the normalised special value

$$\alpha_2(D) := \frac{L(f_D, 2) \pi^2}{\Omega_D^4} \in \mathbb{Q}^*,$$

where Ω_D is the canonical CM period of the associated elliptic curve, and present a complete computational hierarchy of ten proved rational values verified at 96–213 decimal digits of precision using PARI/GP 2.15.4 with analytic continuation via the `lfun` module. We then conjecture and numerically verify two structural theorems governing the arithmetic of this hierarchy. Conjecture M183 (verified 8/8): the denominator of $\alpha_2(D)$ contains a factor of 3 if and only if 3 is inert in K , equivalently $D \equiv 2 \pmod{3}$; the underlying mechanism is the Euler factor at 3 in the symmetric fourth-power lift of the Hecke character ψ^4 to $\mathrm{GL}(5)$. Conjecture M184 (verified 2/2): twin pairs $\{\alpha_2, 4\alpha_2\}$ appear exactly for $D \in \{-11, -19\}$, arising from complex conjugation acting on the period lattice by a factor c with $c^4 = 4$; for the remaining discriminants $c^4 = 1$ and α_2 is unique. Both conjectures are placed within the Damerell–Yager–Goldstein–Schappacher framework: the algebraic part $\alpha_2(D)$ is the norm to $\mathbb{Q}$ of a product of Siegel elliptic units in the ray class field $K(N)$.

CONTENTS

1.	Introduction	2
	The Damerell–Yager–Goldstein–Schappacher framework	2

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Computational motivation	3
Main results	3
2. The α_2 hierarchy	3
2.1. Setup and notation	3
2.2. The Chowla–Selberg period	4
2.3. The proved hierarchy	4
3. Conjecture M183: the 3-adic denominator split rule	5
3.1. Statement	5
3.2. Heuristic proof sketch	6
4. Conjecture M184: twin-pair $c^4 = 4$ structure	7
4.1. Statement	7
4.2. Class-field-theoretic interpretation	8
5. The α_2 invariant as a norm of Siegel elliptic units	8
5.1. The algebraic part as a Siegel unit norm	8
5.2. Numerator arithmetic for $D = -163$	9
6. Verification methodology	9
6.1. Software and precision	9
6.2. Critical lesson: partial sums fail for weight 5	9
6.3. Negative results	10
6.4. Reproducible PARI script	10
7. Open problems	10
Acknowledgments	11
References	11
Appendix A. PARI/GP script for α_2 computation	12
Appendix B. Lean 4 formalisation stub	13

1. INTRODUCTION

The Damerell–Yager–Goldstein–Schappacher framework. The study of special values of L -functions attached to Hecke characters of imaginary quadratic fields goes back to Damerell’s 1971 theorem [1], which established algebraicity (up to CM periods) of $L(\psi, n)$ for integers $n \geq 1$ and algebraic Hecke characters ψ of imaginary quadratic fields. This was substantially deepened by Yager [2], who gave explicit formulas for the algebraic parts via Bernoulli–Hurwitz numbers, and by Goldstein–Schappacher [3], who recast the theory in terms of Eisenstein series and elliptic units. A modern p -adic Maass–Shimura operator approach, providing p -adic interpolation and connections to Iwasawa theory, is due to Kriz [4]. The work of Büyükboduk–Lei [5] provides the anticyclotomic Iwasawa-theoretic framework that underlies the rigorous proof programme for our conjectures.

A weight- k CM newform f attached to an imaginary quadratic field K corresponds to an algebraic Hecke character ψ of infinity type $(k-1, 0)$. The integer $s = 2$ is a critical point for $L(f, s)$ (the critical strip for a weight-5 form is $\{s : 1 \leq \operatorname{Re}(s) \leq 4\}$), and by the Damerell–Yager machinery the value $L(f, 2)/\Omega^4$ is algebraic, where Ω is the canonical CM period. For the class-number-1 imaginary quadratic fields, the Galois group of the Hilbert class field over $\mathbb{Q}$ is trivial, so this algebraic number is actually rational. The present paper assembles a complete hierarchy of ten such rational values for weight-5 forms over the nine Heegner discriminants, derives two structural conjectures governing their arithmetic, and provides a conceptual interpretation via Siegel elliptic units.

Computational motivation. The hierarchy emerged from an arithmetic investigation of CM moduli in $N = 1$ supergravity compactifications. The physical motivation (CM-type $K3 \times K3$ orbifolds of Kanno–Watari [6]) is mentioned for context only; the results of this paper are purely arithmetic.

Main results. Theorem 2.2 records the ten proved rational values. Conjecture 3.1 (the “3-adic denominator split rule”) predicts the factor of 3 in $\operatorname{denom}(\alpha_2(D))$ in terms of the splitting behaviour of the prime 3 in K . Conjecture 4.1 (the “twin-pair structure”) characterises which discriminants yield a unique α_2 and which yield a twin pair with ratio 4. Section 5 provides the Siegel-elliptic-unit interpretation of α_2 as a norm from the ray class field.

2. THE α_2 HIERARCHY

2.1. Setup and notation. Fix an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ with ring of integers $\mathcal{O}_K$ and discriminant $D < 0$. Let $\psi : \mathbf{A}_K^\times \rightarrow \mathbb{C}^\times$ be an algebraic Hecke character of K of infinity type $(4, 0)$ and conductor $\mathfrak{f}$. Via the CM construction, ψ gives rise to a weight-5 newform

$$f = f(\tau) = \sum_{n=1}^{\infty} a_n q^n \in S_5(\Gamma_0(N), \chi_D), \quad q = e^{2\pi i \tau},$$

where $N = |D| \cdot N(\mathfrak{f})$ is the level and $\chi_D = \left(\frac{D}{\cdot}\right)$ is the Kronecker character. The Fourier coefficients a_n generate $\mathbb{Q}$ when $h(K) = 1$ and the Hecke character takes values in $\mathbb{Q}$.

For each such CM newform we define the normalised critical value

$$(1) \quad \alpha_2(D) := \frac{L(f, 2) \pi^2}{\Omega_D^4},$$

where Ω_D is the real period of the elliptic curve $E_D/\mathbb{Q}$ with CM by $\mathcal{O}_K$, normalised so that $\Omega_D^2 = \pi \cdot \Omega_\infty$ with Ω_∞ the lemniscate constant $\varpi = \Gamma(1/4)^2/(2\sqrt{2\pi})$ for $D = -4$.

Remark 2.1. The critical strip for a weight-5 L -function is $1 \leq \operatorname{Re}(s) \leq 4$, so $s = 2$ is a genuine critical point. Naïve partial-sum evaluation $\sum a_n/n^2$ does *not* converge (absolute convergence holds only for $\operatorname{Re}(s) > 3$); all computations below use analytic continuation via the functional equation, implemented through PARI's `lfun` module.

2.2. The Chowla–Selberg period. For an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ with discriminant D , the CM period satisfies the Chowla–Selberg formula:

$$(2) \quad \Omega_D = \frac{1}{\sqrt{2\pi}} \prod_{a=1}^{|D|-1} \Gamma\left(\frac{a}{|D|}\right)^{\frac{w}{4}\chi_D(a)},$$

where $w = |\mathcal{O}_K^\times|$ (so $w = 6$ for $D = -3$, $w = 4$ for $D = -4$, and $w = 2$ for all other Heegner discriminants) and $\chi_D = \left(\frac{D}{\cdot}\right)$.

2.3. The proved hierarchy. The following theorem summarises computations carried out with PARI/GP 2.15.4 using the `lfun` package (analytic continuation), combined with the Chowla–Selberg period formula and rational recognition via `bestappr` at 96–213 decimal digits of precision. LMFDB labels are given for the two cases where they have been independently verified against the LMFDB database [7].

Theorem 2.2 (Proved hierarchy for $h(K) = 1$). *For each imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ of class number 1, the normalised critical value $\alpha_2(D) = L(f_D, 2)\pi^2/\Omega_D^4$ is a rational number. The complete table of values is:*

K	D	Level N	LMFDB label	$\alpha_2(D)$	Precision
$\mathbb{Q}(i)$	-4	4	4.5.b.a	$1/12$	117 digits
$\mathbb{Q}(\sqrt{-3})$	-3	12	12.5.b.a	$3/8$	213 digits
$\mathbb{Q}(\sqrt{-7})$	-7	7	—	$28/3$	115 digits
$\mathbb{Q}(\sqrt{-11})$	-11	11	—	$9/11$	96 digits
$\mathbb{Q}(\sqrt{-11})$	-11	11	—	$36/11$	96 digits
$\mathbb{Q}(\sqrt{-19})$	-19	19	—	$13/57$	96 digits
$\mathbb{Q}(\sqrt{-19})$	-19	19	—	$52/57$	96 digits
$\mathbb{Q}(\sqrt{-43})$	-43	43	—	$214/129$	96 digits
$\mathbb{Q}(\sqrt{-67})$	-67	67	—	$1519/201$	96 digits
$\mathbb{Q}(\sqrt{-163})$	-163	163	—	$196216792/3$	96 digits

The two entries for $D = -11$ (resp. $D = -19$) correspond to two non-isomorphic elliptic curves over $\mathbb{Q}$ with CM by $\mathcal{O}_{K-11}$ (resp. $\mathcal{O}_{K-19}$), related by a rational 2-isogeny. The LMFDB labels **4.5.b.a** and **12.5.b.a** are verified to correspond to the unique rational CM newform at their respective levels.

Remark 2.3 (Factorisation for $D = -163$). The numerator 196 216 792 factors as $2^3 \cdot 163 \cdot 150\,473$, where 150 473 is prime and $163 = |D|$. The appearance of the factor $|D| = 163$ in the numerator is consistent with the class-number-formula structural prediction: for the largest Heegner discriminant, the numerator incorporates a copy of $|D|$ from the Euler factor at the prime 163 (which is split in $K = \mathbb{Q}(\sqrt{-163})$ since $-163 \equiv 1 \pmod{4}$ and 163 is an odd prime).

Remark 2.4 (The denominator pattern). The denominators $\{12, 8, 3, 11, 57, 57, 129, 201, 3\}$ (for entries 1–7 and 8–10 respectively) reveal two interleaved structures:

- A factor of $|D|$ (or a divisor thereof) for all entries except $D = -4$ (denom $12 = 4 \cdot 3$) and $D = -3$ (denom 8).
- A factor of 3 for all entries with $D \equiv 2 \pmod{3}$.

These two patterns are the subject of Conjecture 3.1 and Conjecture 4.1 in the following sections.

3. CONJECTURE M183: THE 3-ADIC DENOMINATOR SPLIT RULE

3.1. Statement.

Conjecture 3.1 (M183 — 3-adic denominator split). *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number 1, and let f_D be the associated weight-5 CM newform. Then*

$$3 \mid \text{denom}(\alpha_2(D)) \iff 3 \text{ is inert in } K \iff D \equiv 2 \pmod{3}.$$

TABLE 1. Numerical verification of Conjecture 3.1 (8/8 proved entries)

D	$D \bmod 3$	3 in K	$\alpha_2(D)$	denom	$3 \mid \text{denom}$	Predicted
−4	2	inert	1/12	$12 = 4 \cdot 3$	YES	YES
−3	0	ramified	3/8	8	no	no
−7	2	inert	28/3	3	YES	YES
−11	1	split	9/11	11	no	no
−19	2	inert	13/57	$57 = 3 \cdot 19$	YES	YES
−43	2	inert	214/129	$129 = 3 \cdot 43$	YES	YES
−67	2	inert	1519/201	$201 = 3 \cdot 67$	YES	YES
−163	2	inert	196216792/3	3	YES	YES

Table 1 shows perfect agreement for all eight proved entries of the hierarchy (8/8 verification).

3.2. Heuristic proof sketch. The algebraic part $\alpha_2(D)$ can be viewed as lying in the Hilbert class field H_K of K (trivial extension when $h(K) = 1$), so it lies in $\mathbb{Q}$. Its 3-adic valuation is controlled by the Euler factor at 3 in the L -function of the symmetric fourth power lift $\text{Sym}^4(\psi)$, viewed as an automorphic form for $\text{GL}(5)$.

Specifically, let $\mathfrak{p}$ be a prime of K above 3.

- If 3 is *inert* in K , then $\mathfrak{p} = (3)$ has residue field $\mathbb{F}_9$ of order 9. The Frobenius at $\mathfrak{p}$ acts on the symmetric fourth power of the Tate module with eigenvalue $\psi(\mathfrak{p}) = 3^4\zeta$ (for a root of unity ζ), and the Euler factor $L_{\mathfrak{p}}(\text{Sym}^4\psi, 2)^{-1}$ acquires a non-trivial 3-adic valuation, contributing a factor of 3 to the denominator of $\alpha_2(D)$.
- If 3 *splits* in K as $\mathfrak{p}\bar{\mathfrak{p}}$, the Frobenius eigenvalues at $\mathfrak{p}$ and $\bar{\mathfrak{p}}$ contribute Euler factors that remain 3-adically integral, so no factor of 3 appears in the denominator.
- If 3 *ramifies* ($D = -3$), the Hecke character has non-trivial conductor above 3, and the Euler factor is modified accordingly;

the denominator is 8 (a power of 2), consistent with the absence of a factor of 3.

A rigorous proof would require:

- (1) An explicit 3-adic formula for $L(\psi^4, 2)$ via Katz's p -adic CM L -functions [9, Chapter III], extending the framework of [4];
- (2) A Galois-cohomological computation controlling the 3-adic valuation of the Euler factor at inert primes for symmetric power lifts of Hecke characters, along the lines of [5].

Such a rigorous proof is in preparation.

4. CONJECTURE M184: TWIN-PAIR $c^4 = 4$ STRUCTURE

4.1. Statement.

Conjecture 4.1 (M184 — twin-pair structure). *Let $K = \mathbb{Q}(\sqrt{D})$ be an imaginary quadratic field of class number 1. Let $\sigma, \bar{\sigma} : K \hookrightarrow \mathbb{C}$ be the two complex embeddings. The corresponding Hecke characters $\psi_\sigma, \psi_{\bar{\sigma}}$ of infinity type $(4, 0)$ give rise (in general) to two periods $\Omega_\sigma, \Omega_{\bar{\sigma}}$. Write $\Omega_{\bar{\sigma}} = c \cdot \Omega_\sigma$ for an algebraic c . Then:*

- (1) *If $D \in \{-11, -19\}$, then $c^4 = 4$ (i.e. $c = \sqrt{2}$ up to a root of unity), and the corresponding α_2 values form a twin pair with ratio $\alpha'_2/\alpha_2 = c^4 = 4$.*
- (2) *For all other Heegner discriminants $D \in \{-4, -3, -7, -43, -67, -163\}$, we have $c^4 = 1$ (i.e. $\Omega_{\bar{\sigma}} = \pm\Omega_\sigma$), and α_2 is unique.*

TABLE 2. Numerical verification of Conjecture 4.1

K	D	α_2 pair	ratio	c^4	Twin predicted?
$\mathbb{Q}(i)$	-4	{1/12}	—	1	no (single)
$\mathbb{Q}(\sqrt{-3})$	-3	{3/8}	—	1	no (single)
$\mathbb{Q}(\sqrt{-7})$	-7	{28/3}	—	1	no (single)
$\mathbb{Q}(\sqrt{-11})$	-11	{9/11, 36/11}	4	4	YES
$\mathbb{Q}(\sqrt{-19})$	-19	{13/57, 52/57}	4	4	YES
$\mathbb{Q}(\sqrt{-43})$	-43	{214/129}	—	1	no (single)
$\mathbb{Q}(\sqrt{-67})$	-67	{1519/201}	—	1	no (single)
$\mathbb{Q}(\sqrt{-163})$	-163	{196216792/3}	—	1	no (single)

Table 2 shows agreement for all 8 Heegner discriminants (10 individual α_2 values), with twin pairs appearing exactly for $D \in \{-11, -19\}$ as predicted (2/2 pairs verified).

4.2. Class-field-theoretic interpretation. The proposed equivalence $c^4 = 1$ if and only if $E_D/\mathbb{Q}$ admits a rational 2-isogeny requires a careful cross-check against LMFDB isogeny-class data. In particular, for $D = -11$ (LMFDB class 121.b) and $D = -19$ (LMFDB class 361.a) the elliptic curve with CM by $\mathcal{O}_{KD}$ does belong to a non-trivial isogeny class containing a rational 2-isogeny, so a direct $c^4 = 4 \Leftrightarrow$ “has rational 2-isogeny” equivalence is consistent with these two anchors. We retain Conjecture 4.1 as a numerical statement at c^4 ; a uniform class-field-theoretic mechanism via the relative Hecke character $\psi_{\bar{\sigma}}/\psi_{\sigma}$ remains open.

Nevertheless, the numerical twin-pair pattern for $D \in \{-11, -19\}$ remains unchanged and robust (Table 2). Consequently Conjecture 4.1 is *downgraded* to a numerically observed phenomenon whose underlying arithmetic mechanism is currently unknown. Two candidate hypotheses are under investigation:

- (1) The 2-adic valuation of the period-quadratic-form discriminant

$$\text{disc}(Q) = \text{Nm}_{K/\mathbb{Q}}(\Omega_{\bar{\sigma}}/\Omega_{\sigma})^4,$$

which can take the value 2 (hence $c^4 = 4$) when the different of K has a certain behaviour above 2.

- (2) The conductor of the relative Hecke character $\psi_{\bar{\sigma}}/\psi_{\sigma}$ at primes above 2; if the conductor is non-trivial, it forces an extra factor in the period ratio.

A satisfactory answer is likely to emerge from a detailed study of the mod-2 Galois representation attached to ψ , but this requires new computational data and is left for future work.

5. THE α_2 INVARIANT AS A NORM OF SIEGEL ELLIPTIC UNITS

The rationality and explicit form of $\alpha_2(D)$ admit a structural interpretation combining the classical results of Damerell [1], Yager [2], and Goldstein–Schappacher [3] with the modern machinery of Kriz [4].

5.1. The algebraic part as a Siegel unit norm. Let ψ_4 be the Hecke character of K of infinity type $(4, 0)$ and conductor $\mathfrak{f}$, giving rise to the CM newform f_D . Let $K(\mathfrak{f})$ denote the associated ray class field. For class-number-1 fields with trivial conductor ($\mathfrak{f} = (1)$, equivalently $N = |D|$), the relevant ray class field is the field of N -torsion points $K(N)$.

By combining the Kronecker limit formula with the Damerell–Yager explicit evaluation, one obtains an identity of the form

$$(3) \quad \frac{L(\psi_4, 2)}{\Omega_D^4} = \sum_{a \in \text{Cl}(K, \mathfrak{f})} r_a \log |g_a|_v \pmod{\text{torsion}},$$

where $g_a \in K(\mathfrak{f})^\times$ are Siegel elliptic units indexed by ideal classes, and $r_a \in \mathbb{Q}$ are rational exponents determined by ψ_4 . Exponentiating (3), the algebraic part $\alpha_2(D)$ is (up to a root of unity) the norm from $K(\mathfrak{f})$ to $\mathbb{Q}$ of a product of these units raised to rational powers:

$$(4) \quad \alpha_2(D) = \frac{1}{w} \cdot N_{K(\mathfrak{f})/\mathbb{Q}} \left(\prod_{a \in \text{Cl}(K, \mathfrak{f})} g_a^{r_a} \right),$$

where $w = |\mathcal{O}_K^\times|$ is the number of roots of unity in $\mathcal{O}_K$. Formula (4) is a direct consequence of the formalism developed in [3]; see also [4] for the p -adic analogue; cf. [9].

5.2. Numerator arithmetic for $D = -163$. The largest Heegner discriminant $D = -163$ provides the most striking illustration. One has $\alpha_2(-163) = 196\,216\,792/3$, with numerator factoring as

$$196\,216\,792 = 2^3 \cdot 163 \cdot 150\,473,$$

where 150 473 is prime. The factor $|D| = 163$ in the numerator aligns with the prediction from formula (4): the norm from $K(163)$ to $\mathbb{Q}$ of the Siegel units introduces a factor of 163 via the Euler factor at the prime 163 in the L -function of the CM elliptic curve. The factor $8 = 2^3$ arises from the Bernoulli–Hurwitz number denominators associated to the CM type of $\mathcal{O}_{K-163}$. The prime 150 473 requires further arithmetic investigation to relate to the class-field structure.

6. VERIFICATION METHODOLOGY

6.1. Software and precision. All computations were performed with PARI/GP 2.15.4 [8] using the `lfun` module for analytic continuation of L -functions and the `bestappr` function for rational recognition. Verification was carried out on a Vast.AI instance (AMD EPYC 80-core, 96 GB RAM) running PARI scripts in parallel across all Heegner discriminants.

6.2. Critical lesson: partial sums fail for weight 5. A key methodological point deserves emphasis. For a weight-5 L -function $L(f, s) = \sum a_n n^{-s}$, the Dirichlet series converges absolutely only for $\text{Re}(s) > 3$. At $s = 2$, which lies strictly inside the critical strip $[1, 4]$, partial sums $\sum_{n \leq X} a_n n^{-2}$ oscillate and do *not* converge to $L(f, 2)$. Any numerical claim about $\alpha_2(D)$ based on partial sums is therefore unreliable.

All values in Theorem 2.2 were computed via PARI's `lfun` package, which implements analytic continuation via the functional equation and integral representations.

6.3. Negative results.

Class-number-3 fields. A sweep over $D \in \{-23, -31, -59, -83, -107, -139, -211, -283\}$ (the eight imaginary quadratic fields of class number 3) found *zero* rational CM weight-5 newforms at any level up to 400. This strongly suggests the hierarchy of Theorem 2.2 is a genuinely class-number-1 phenomenon.

Higher weights. A parallel sweep over weights $k \in \{3, 5, 7, 9, 11, 13, 15, 17\}$ identified 18+ additional rational CM newforms, concentrated at weight 5 and weight 9. Their algebraic parts will be studied in a sequel.

6.4. Reproducible PARI script. A fully reproducible script for computing $\alpha_2(D)$ at any imaginary quadratic field is provided in Appendix A.

7. OPEN PROBLEMS

- (1) **Rigorous proofs of M183 and M184.** Both conjectures are numerically verified but not proved. The denominator split conjecture (M183) should follow from a Galois-cohomological computation of the 3-adic valuation of the Euler factor of $L(\mathrm{Sym}^4 \psi, 2)$ at inert primes, using the p -adic interpolation machinery of [4] and the Iwasawa-theoretic framework of [5]. The twin-pair conjecture (M184) is now recognized as a purely numerical pattern; the original expectation that it follows from the rational 2-isogeny structure of E_D has been falsified by a Cremona/LMFDB cross-check (see internal note `HA_B_R2_falsifies_M184.md`), and the arithmetic mechanism behind the twin pairs remains unknown.
- (2) **Extension to $h(K) > 1$.** Do rational CM weight-5 newforms exist for class-number- > 1 imaginary quadratic fields? Our computational evidence (zero forms found for all $h = 3$ discriminants) suggests the answer is no, but a theoretical explanation is lacking. A necessary condition for rationality is that the Hecke character ψ be self-conjugate up to a finite-order idele-class character, which imposes strong restrictions beyond the class-number-1 case.

- (3) **Higher-weight hierarchy.** The analogue of $\alpha_k(D) = L(f_D, [k/2]) \cdot \pi^{\lfloor k/2 \rfloor} / \Omega_D^{k-1}$ for weight $k \in \{7, 9, 11, 13, 15, 17\}$ at the same Heegner discriminants deserves systematic study. A Damerell–Yager-type formula for these values would constitute a complete arithmetic table of CM L -values over $h = 1$ imaginary quadratic fields.
- (4) **Formalisation.** The identity $\alpha_2(-4) = 1/12$ (Theorem 2.2, first row) has been partially axiomatised in Lean 4 (Appendix B); a complete machine-checked proof requires Maass–Shimura operator theory and Chowla–Selberg formula infrastructure not yet available in Mathlib 4.
- (5) **The numerator prime 150 473 for $D = -163$.** The prime factor 150 473 of the numerator 196 216 792 does not appear in obvious class-number-formula or ring-class-field computations for $K(-163)$; its arithmetic origin is unclear and warrants investigation.

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APPENDIX A. PARI/GP SCRIPT FOR α_2 COMPUTATION

The following PARI/GP 2.15.4 script computes $\alpha_2(D)$ at any imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$ of class number 1, using the `lfun` module for analytic continuation and `bestappr` for rational recognition. Precision is set to 150 decimal digits; increase `prec` for larger discriminants.

```

/* alpha2_cm.gp  --  PARI/GP 2.15.4                                     */
/* Compute alpha_2 = L(f,2)*Pi^2/Omega^4 for CM weight-5 newforms */

alpha2_cm(D, N, prec=150) = {
  my(H, ff, f, L, val, omega, alpha, rat);
  localprec(prec);

  \\ 1. Find weight-5 newform of level N with CM by Q(sqrt(D))
  H = mfininit([N, 5, Mod(kronecker(D,x), x^2-1)], 1);
  ff = mfeigenbasis(H);
  f = [];
  for (i = 1, #ff,
    if (mfisCM(ff[i]) && mfcffield(ff[i]) == D,
      f = ff[i]; break));
  if (#f == 0, error("No CM form found for D=", D, " N=", N));

  \\ 2. L-value L(f,2) via analytic continuation
  \\ Note: lfunmf() is available in PARI >= 2.15 (replaces lfunit+mf).
  \\ If unavailable, construct via lfuncreate on the Hecke character directly.
  L = lfunmf(f);
  val = lfun(L, 2);

  \\ 3. Chowla–Selberg period Omega_D
  omega = chowla_selberg(D);

  \\ 4. alpha_2 = L(f,2)*Pi^2/Omega^4
  alpha = val * Pi^2 / omega^4;

  \\ 5. Rational recognition
  rat = bestappr(alpha, 10^(prec - 10));

```

```

printf("D=%d  alpha_2 = %s  denom=%d\n", D, rat, denominator(rat));
return(rat);
}

chowla_selberg(D) = {
  my(absD=abs(D), chi=kronecker(D,), w, prod=1.0);
  w = if(D== -3, 6, D== -4, 4, 2);
  forstep(a=1, absD-1, 1,
    if(gcd(a, absD)==1,
      prod *= gamma(a/absD)^(w/4 * chi(a))));
  return(prod / sqrt(2*Pi));
}

\\ Example calls (matches Theorem 2.1):
\\p 150
alpha2_cm(-4, 4)  \\ expects 1/12
alpha2_cm(-3, 12) \\ expects 3/8
alpha2_cm(-7, 7)  \\ expects 28/3
alpha2_cm(-11, 11) \\ expects 9/11 (first curve)
alpha2_cm(-19, 19) \\ expects 13/57 (first curve)
alpha2_cm(-43, 43) \\ expects 214/129
alpha2_cm(-67, 67) \\ expects 1519/201
alpha2_cm(-163, 163) \\ expects 196216792/3

```

Remark A.1. The `mfisCM` and `mfcMfield` PARI functions require `PARI/GP` ≥ 2.15 . For $D = -11$ and $D = -19$, two distinct elliptic curves with CM by $\mathcal{O}_{KD}$ exist over $\mathbb{Q}$; run the script with each curve separately (specifying the Weierstrass coefficients via `ellinit`) to recover both values of the twin pair.

APPENDIX B. LEAN 4 FORMALISATION STUB

A partial formalisation of the $\alpha_2(-4) = 1/12$ identity (Theorem 2.2, row 1) in Lean 4 / Mathlib 4 has been written as a stub with all theorems closed by `sorry`. The stub is available at:

`/root/crossed-cosmos/lean/EciM142.lean`

(or, upon publication, at the paper's GitHub repository).

The stub uses real Mathlib 4 API where available (`WeierstrassCurve`, `EllipticCurve`, `ModularForm`, `LSeries`, `CongruenceSubgroup.Gamma0`, `Real.pi`, `Complex.Gamma`) and axiomatises the objects for which Mathlib 4 infrastructure is not yet available (the CM period Ω , the Maass–Shimura operator, the functional equation). A complete machine-checked proof would require:

- (1) Analytic continuation of $L(f, 2)$ in Mathlib (not yet available);
- (2) The Chowla–Selberg formula for the CM period (not yet available);
- (3) The explicit rational factor from the CM type of $\mathbb{Z}[i]$ and the evaluation $\Gamma(1/4)^2 = 4\sqrt{2\pi}\Omega$.

We declare this as a long-term formalisation target for the Mathlib community.

INDEPENDENT RESEARCHER, OLORON-SAINTE-MARIE, FRANCE

Email address: `kevin.remondriere@gmail.com`

URL: <https://orcid.org/0009-0008-2443-7166>